

Software:

ASE (Atomic Simulation Environment)

- Structure preparation
- Calculations and optimizations
- Graphical User Interface
- <https://ccml-uconn.github.io/CCML/ASE/>

LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator)

- classical molecular dynamics/material simulations
- <https://github.com/lammps/lammps>
- <https://www.lammps.org/tutorials.html>

OVITO (Scientific data visualization and analysis solution for particle-based simulations)

- Python packages -> usage in notebooks
- https://www.ovito.org/?utm_source=chatgpt.com

Why not AI-/CGI

- It's not a scientific Analysis/Visualization tool as OVITO

Protocol definition

Protocol section	Definition
Research Question	What should be tested?
Material classification	What kind of MOF?
Structural preparation	Document CIF file, cleaning, supercell, charges, and boundary conditions
Model assumptions	Force field, adsorbate model, rigid/flexible framework, temperature, pressure range
Reference data	Literature values or experimental data for comparison
Sampling strategy	Pressure points, simulation duration, number of iterations
Simulation settings	Cutoff, electrostatics, number of steps, software version
Evaluation	Isotherm, error, runtime, uncertainty (if applicable)

Scope of validity	For which pressure range is the model plausible?
Reproducibility	Archive input files, scripts, versions, and tables

Module	Metric	What it measures	Possible unit / output	Use in benchmark
A: Baseline GCMC / Adsorption	CO ₂ uptake	Amount of CO ₂ adsorbed at each pressure point	mmol/g or mol/kg	Main simulation output
A	Adsorption isotherm	CO ₂ uptake as a function of pressure	plot / curve	Checks whether the adsorption trend is reproduced
A	Absolute / excess adsorption	Difference between simulated absolute loading and experimentally comparable excess loading	mmol/g or mol/kg	Ensures correct comparison with reference data
A	Error to reference	Deviation between simulated and literature/reference uptake	MAE, relative error, MRE (%)	Main accuracy criterion
A	Runtime	Computational cost per pressure point or complete isotherm	s, min, h	Measures practical efficiency
A	Scope of application / validity range	Pressure and system range where the model remains reliable	pressure range, material class	Defines where the benchmark result is valid
B: Mixture adsorption / Active Learning	Component uptake	Adsorbed amount of each gas component	mmol/g or mol/kg	Needed for gas mixtures
B	CO ₂ /N ₂ selectivity	Preference of the MOF for CO ₂ over N ₂	dimensionless	Main separation metric
B	MRE	Mean relative error of surrogate or reduced sampling compared to full GCMC	%	Accuracy of active-learning approach
B	R ²	Agreement between predicted and reference adsorption values	dimensionless, 0–1	Quality of surrogate model

B	Required GCMC points	Number of simulations needed to reach target accuracy	number of points	Measures sampling effort
B	Saved simulation points	Reduction compared with full grid sampling	number or %	Measures efficiency gain
B	Uncertainty / PAC	Model uncertainty or perceived accuracy criterion	uncertainty value / PAC	Determines active-learning convergence
C: Potential Benchmark	Isotherm deviation	Difference between isotherms from different potentials	MAE, MRE, curve deviation	Measures influence of force field or MLIP
C	Heat of adsorption / qst	Strength of adsorbate-framework interaction	kJ/mol	Evaluates interaction strength
C	Widom insertion energy	Energy of inserting a test molecule into the MOF	kJ/mol	Tests local adsorption energetics
C	Henry region	Low-pressure adsorption behaviour	Henry constant, low-pressure slope	Tests potential quality at low loading
C	Error to reference	Deviation from experimental, DFT, or fine-tuned reference data	MAE, MRE (%)	Main validation metric
C	Runtime per step	Computational cost of each potential model	s/step or steps/s	Compares efficiency of potentials
C	Systematic over-/underestimation	Whether the model consistently predicts too high or too low adsorption	qualitative / bias value	Identifies model bias
D: Flexibility Benchmark	Rigid vs. flexible isotherm difference	Change in adsorption when framework flexibility is included	MAE, MRE, curve difference	Tests importance of flexibility
D	Structural changes	Changes in framework geometry during simulation	bond lengths, angles, cell parameters	Detects framework response
D	Pore volume changes	Change in accessible pore volume	Å ³ , cm ³ /g, %	Links structure change to adsorption
D	RMSD / geometric deviation	Overall structural deviation from initial/reference structure	Å	Quantifies framework movement

D	Adsorption change due to flexibility	Difference in uptake caused by flexible modelling	mmol/g, %	Measures flexibility effect
D	Convergence over relaxation cycles	Stability of uptake and structure over repeated cycles	change per cycle, %	Checks convergence of coupled workflow

Module	Main benchmark question	Metrics
A: Baseline GCMC	Can a classical GCMC workflow reproduce CO ₂ adsorption trends?	CO ₂ uptake, adsorption isotherm, absolute/excess adsorption, error to reference, runtime, validity range
B: Mixture adsorption / Active Learning	How accurately and efficiently can mixture adsorption landscapes be sampled?	Component uptake, CO ₂ /N ₂ selectivity, MRE, R ² , required GCMC points, saved simulation points, uncertainty/PAC
C: Potential Benchmark	How strongly do force fields or MLIPs influence adsorption predictions?	Isotherm deviation, heat of adsorption, Widom insertion energy, Henry region, error to reference, runtime per step, systematic over-/underestimation
D: Flexibility Benchmark	Does framework flexibility significantly affect adsorption behaviour?	Rigid vs. flexible isotherm difference, structural changes, pore volume changes, RMSD/geometric deviation, adsorption change due to flexibility, convergence over relaxation cycles

Possible MOF-5 sources:

WMD-group MOF-5: https://github.com/WMD-group/Crystal_structures/blob/master/MOFs/MOF-5/MOF5.cif

CoRE MOF: <https://github.com/coudertlab/CoRE-MOF> (MOF-5, IRMOF-1, SAHYIK, CCDC 256965, Zn₄O(BDC)₃)

CSD/CCDC: <https://www.ccdc.cam.ac.uk/support-and-resources/downloads/> (need for request) als CoRE-MOF available there (modified & unmodified)

Force fields sources:

Force fields, CIF files +. Input files: <https://zenodo.org/records/8190237> (preferred)

<https://github.com/iRASP/RASPA2/tree/master/forcefield>

<https://github.com/coudertlab/simple-adsorption-workflow/tree/main/parameters/forcefield>

For LAMMPS: <https://github.com/MOFplus/cmc-tools>

Regence Data:

<https://zenodo.org/records/819023> (source <https://pubs.acs.org/doi/10.1021/acs.jced.2c00583>)

<https://mof.tech.northwestern.edu/databases> (web-based catalog of adsorbent materials and measured adsorption properties of numerous materials obtained from article entries from the scientific literature)

<https://github.com/NIST-ISODB/isodb-library>

Modul	Ziel	LAMMPS-Methode	Output	Python-Auswertung
A	CO ₂ -Isotherme	fix gcmc für CO ₂	Molekülzahl, Energie, Laufzeit	Uptake, Isotherme, Fehler

Modul	Ziel	LAMMPS-Methode	Output	Python-Auswertung
B	CO ₂ /N ₂ -Mischung	GCMC für mehrere Molekülsorten	N_CO ₂ , N_N ₂	Selectivity, MRE, R ²
C	Henry/Widom	<code>fix widom</code>	Insertionsenergie, Henry-Konstante	Henry-Bereich, Abweichung
D	Flexibilität	Minimierung, MD, ggf. GCMC+MD	Trajektorien, Struktur, Energie	RMSD, Porenänderung, flexible vs. rigid

Schritt	Aufgabe	Was ändert sich durch LAMMPS-only?
1. MOF-5-Struktur vorbereiten	CIF laden, Unit Cell prüfen, Supercell erzeugen	ASE/Python bleibt sinnvoll, aber du musst am Ende ein korrektes LAMMPS data file erzeugen. LAMMPS nutzt für komplexe Systeme üblicherweise Data-Files, die mit <code>read_data</code> eingelesen werden. (LAMMPS Dokumentation)
2. Atomtypen und Charges definieren	MOF-Atome typisieren, Partialladungen setzen	Du musst Charges, Masses, Pair Coeffs usw. explizit in LAMMPS-kompatibler Form bereitstellen.
3. Adsorbat-Moleküle vorbereiten	CO ₂ , N ₂ usw. als Molekül-Templates	Für GCMC mit Molekülen brauchst du LAMMPS-molecule-Templates, z. B. für CO ₂ oder N ₂ . (LAMMPS Dokumentation)
4. Force Field definieren	Framework–Framework, Gas–Gas, Framework–Gas	In RASPA wäre vieles adsorptionstypisch vorbereitet. In LAMMPS musst du Pair Styles, Mixing Rules, Cutoffs, Coulomb-Behandlung und Molekülparameter sauber selbst setzen.
5. Simulation pro Modul starten	GCMC, Widom, MD, Relaxation	Jedes Modul wird ein eigenes LAMMPS-Inputskript oder ein Template mit Variablen.
6. Output auswerten	Uptake, Isotherme, Selectivity, RMSD, Laufzeit	Python liest <code>log.lammps</code> , Dumps, Molekülzahlen, Energien usw. und berechnet daraus deine Metriken.
7. Benchmark-Report	Tabellen, Plots, Vergleich zu Referenzdaten	Die Metriken bleiben fast gleich, aber du dokumentierst jetzt LAMMPS-Parameter sehr genau.

A:

MOF-5 CIF

→ LAMMPS data file

→ CO₂ molecule template

→ LAMMPS GCMC bei mehreren Druckpunkten

→ mittlere CO₂-Molekülzahl im Porenraum

→ Umrechnung in uptake, z. B. mol/kg oder cm³(STP)/g

→ Vergleich mit Referenzisotherme

Metrik	Bleibt?	Änderung
CO ₂ uptake	Ja	Muss aus der durchschnittlichen Molekülzahl in LAMMPS berechnet werden
Adsorptionsisotherme	Ja	Druckpunkte werden über chemisches Potential/Fugazität abgebildet
Absolute/excess adsorption	Ja	Umrechnung musst du selbst in Python machen
Error to reference	Ja	Vergleich gegen Literatur/RASPA/CRAFTED-Daten
Runtime	Ja	LAMMPS-Laufzeit pro Druckpunkt messen
Scope of application	Ja	Jetzt: „LAMMPS-only GCMC für MOF-5“

B:

MOF-5 data file

→ CO₂ molecule template

→ N₂ molecule template

→ mehrere GCMC-Fixes oder sequentielle GCMC-Schritte

→ mittlere Anzahl CO₂ und N₂

→ component uptake

→ Selectivity berechnen

Metrik	Wie mit LAMMPS berechnen?
Component uptake	Durchschnittliche Anzahl von CO ₂ - und N ₂ -Molekülen aus LAMMPS-Output
CO ₂ /N ₂ -Selectivity	Aus adsorbierter Zusammensetzung und Gasphasen-Zusammensetzung
MRE	Fehler gegenüber Referenzdaten
R ²	Fit/Vergleich zwischen vorhergesagter und Referenzisotherme
Number needed GCMC-points	Anzahl tatsächlich simulierter Druckpunkte
Saved simulation points	Vergleich zu vollständigem Referenzraster
Uncertainty/PAC	Kommt eher aus deinem Python/ML-Auswertungsteil, nicht direkt aus LAMMPS

C:

MOF-5 data file

→ CO₂ molecule template

→ Widom-Insertionen mit LAMMPS

→ Henry-Konstante / Insertionsenergie

→ Vergleich mit GCMC-Isotherme im niedrigen Druckbereich

→ Fehleranalyse

Metrik	Umsetzung mit LAMMPS
Deviation of isotherms	GCMC-Isotherme vs. Widom/Henry-Näherung
Heat of adsorption	Aus Energieauswertung oder separater post-processing Berechnung
Widom insertion energy	Direkt über Widom-Sampling ableitbar
Henry-area	Niederdruckbereich der Isotherme

Metrik	Umsetzung mit LAMMPS
Error to reference	Vergleich zu Literatur/RASPA/CRAFTED
Runtime per step	LAMMPS-Logdateien auswerten
Systematic over-/underestimation	Residuenplot oder Fehlertrend über Druckpunkte

D:

MOF-5 initial

- Energie-Minimierung
- optional NVT/NPT-MD
- flexible Framework Simulation
- Adsorption mit starrer Struktur vergleichen
- Strukturänderungen analysieren

Metrik	Umsetzung mit LAMMPS
Difference rigid vs. flexible isotherm	GCMC mit fixiertem Framework vs. GCMC/MD mit flexiblem Framework
Structure changes	Dump-Dateien analysieren
Pore volume changes	Extern mit Python/Porenanalyse-Tool oder geometrischer Näherung
RMSD/geometric changes	Python/ASE/MDAnalysis
Adsorption changes through flexibility	Uptake flexibel minus uptake rigid
Convergence over relaxation cycles	Energie, RMSD, Zellparameter über Zyklen